

Automated Heartbeat Classification from ECG Signals

Using a 1D Convolutional Neural Network, with machine learning baselines and ensemble learning
· Project Blueprint

1. Problem Statement

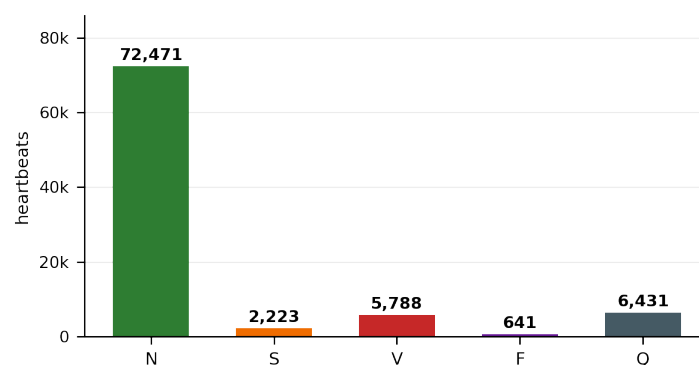
Cardiovascular disease is a leading cause of death worldwide, and the electrocardiogram (ECG) is the standard tool for detecting abnormal heart rhythms. A single long-term recording can contain tens of thousands of heartbeats, while abnormal beats may occur only occasionally. Reviewing them manually is slow and depends on expert availability.

Core challenge: Given one segmented ECG heartbeat, automatically assign it to a normal or abnormal rhythm category, with reliable performance on the rare abnormal classes rather than high overall accuracy alone.

2. Dataset and Classes

The project uses the **MIT-BIH Arrhythmia Database** (PhysioNet): 48 half-hour ambulatory ECG recordings in which every heartbeat has been annotated by cardiologists. The original beat labels are grouped into the five standard AAMI categories.

Code	Category	Example beats	Share of data
N	Normal	Normal beat, bundle branch block	≈ 83%
S	Supraventricular ectopic	Atrial premature beat	≈ 2.5%
V	Ventricular ectopic	Premature ventricular contraction	≈ 6.6%
F	Fusion	Fusion of ventricular and normal	≈ 0.7%
Q	Unclassifiable	Paced or unknown beat	≈ 7.3%



Class distribution in the training set.

Class imbalance: Normal beats outnumber fusion beats by roughly 113 to 1. A model that always answered “normal” would score about 83% accuracy while being useless. This drives both the training strategy (class weighting) and the choice of evaluation metrics in Section 6.

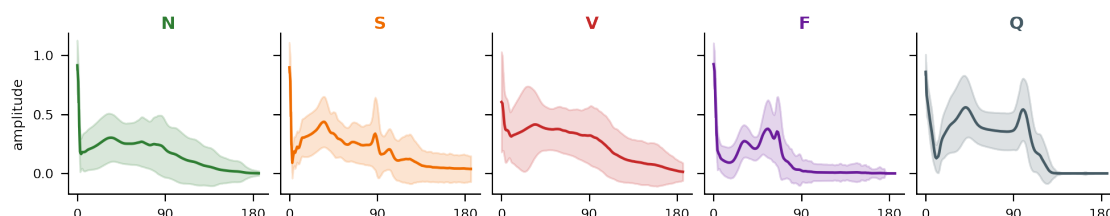


Figure 1 — Average heartbeat shape for each class (shaded band = one standard deviation). Class V is clearly distinct; class S closely resembles N.

3. Clinical Guidance

Guidance: The project is carried out with guidance from a practising medical doctor, who advises on the definitions of the arrhythmia classes, helps interpret the ECG waveforms, and reviews how the system is described so that its scope is stated responsibly.

“The purpose is to develop an educational AI-based tool that automatically classifies ECG heartbeat patterns. It is not a medical device, it has not been clinically validated, and it is not intended to replace professional medical diagnosis.”

4. Why a 1D CNN

An ECG is a one-dimensional sequence of voltage values over time, so a 1D CNN operates on it directly.

- **Native format:** no conversion step is required.
- **Learned features:** the filters pick up P, QRS and T wave shapes without manual feature design.
- **Timing preserved:** converting the signal to an image would turn exact sample timing into pixel distances, and timing distinguishes premature beats.
- **Efficient:** ~187 values per beat rather than ~150,000 pixel values.

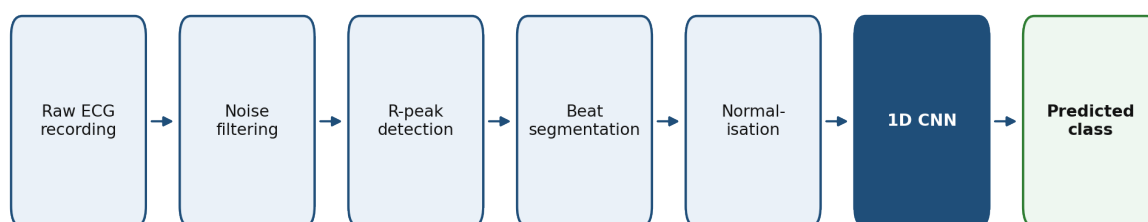


Figure 2 — Processing pipeline, from the raw recording to a predicted arrhythmia class.

5. Model Architecture

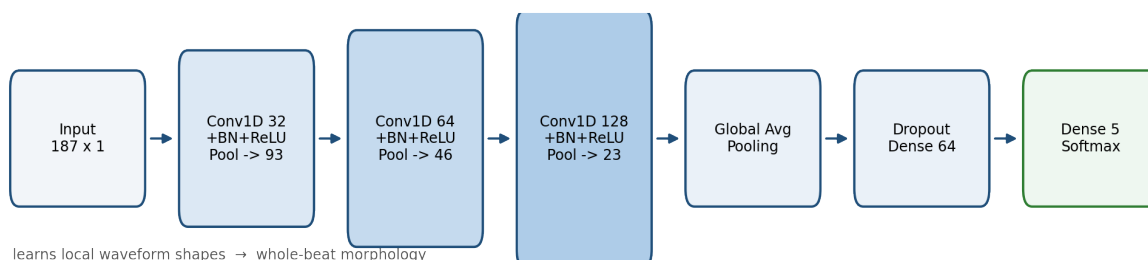


Figure 3 — 1D CNN architecture. Each convolution block halves the sequence length and doubles the filter count.

Stage	Configuration	Purpose
Input	(187, 1)	One normalised heartbeat
Conv block 1	Conv1D 32 filters, kernel 5 + BatchNorm + ReLU + MaxPool 2	Learns small local shapes

Stage	Configuration	Purpose
Conv block 2	Conv1D 64 filters, kernel 5 + BatchNorm + ReLU + MaxPool 2	Combines them into larger patterns
Conv block 3	Conv1D 128 filters, kernel 5 + BatchNorm + ReLU + MaxPool 2	Whole-beat morphology
Pooling	Global average pooling	Compact, overfitting-resistant summary
Classifier	Dropout 0.3 → Dense 64 → Dropout → Dense 5 → Softmax	Probability for each class

Training uses categorical cross-entropy with class weights, the Adam optimiser, and early stopping on validation macro F1.

6. Evaluation and Comparison

The 1D CNN is compared against classical models trained on hand-crafted ECG features (R-R intervals, QRS duration, amplitude and statistical descriptors). Every model is evaluated on an identical data split.

Element	Protocol
Models compared	1D CNN (raw beats) vs Random Forest and XGBoost (hand-crafted features)
Data split	Inter-patient split: beats from one patient appear in either training or testing, never both, so results reflect performance on unseen patients
Imbalance handling	Class weighting in the loss function during training
Primary metric	Macro F1-score — treats all five classes equally
Secondary metrics	Per-class precision and recall, confusion matrix
Accuracy	Reported, but never alone, since it is inflated by the majority class

7. Ensemble Learning

The CNN reads raw waveform shape while the tree-based models read timing and amplitude features, so the two tend to make different errors. Combining them may recover beats that one model alone misses.

- **Soft voting** — average the predicted class probabilities.
- **Weighted voting** — weight each model by its validation macro F1.
- **Stacking** — a small meta-classifier trained on the models' output probabilities.

The meta-learner is trained on validation data only; the ensemble is then evaluated on the same held-out test set as every other model.

8. Expected Outcome and Future Scope

- A trained 1D CNN with documented accuracy, macro F1 and per-class recall, including honest reporting of weak classes such as F.
- A like-for-like comparison against Random Forest and XGBoost, plus an ensemble result.
- A Streamlit interface that returns a predicted class and confidence for an input heartbeat.
- Possible extensions: validation on a second database (for example PTB or INCART), explainability through Grad-CAM, and model compression for wearable hardware.

Research publication intent: The inter-patient evaluation, the controlled comparison between deep and classical models, and the full per-class reporting are the elements a comparative study would need. Once results are complete and validated on a second dataset, this work could be prepared for submission to a student or early-career venue in biomedical signal processing.